MA2040 Jan-Apr 2024
Problem Sheet 4

1. The median of a discrete Random variable X is defined to be the average of the set of all values x such that $P(X \leq x) \geq \frac{1}{2}$ and $P(X \geq x) \geq \frac{1}{2}$. Find the median of a random variable with distriibution $P(X = 1) = \frac{1}{4}, P(X = 2) = \frac{1}{2}, P(X = 3) = \frac{1}{4}$.

2. The median of a continuous random variable X with distribution function F is equal to m scuh that $F(m) = \frac{1}{2}$. Find the median of the random variables with density function
   (a) $f(x) = e^{-1}, X \geq 0$
   (b) $f(x) = 1, 0 \leq x \leq 1$.

3. Show that if $F_{X,Y}$ is a diffenrentiable function at (a, b) then
$$\frac{\partial^2}{\partial x \partial y} F(x, y)|_{(a,b)} = f_{X,Y}(a, b).$$

4. Show that uncorrelated does not, in general, imply independence.

5. Show that if $X, Y \in \{0, 1\}$, then uncorrelated implies independence. How do you interpret this?

6. Find the first four moments of a $N(\mu, \sigma^2)$ random variable.

7. Let Φ be the CDF of a standard normal random variable, that is
$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{\frac{-t^2}{2}} dt.$$
   Show that $\Phi(-z) = 1 - \Phi(z)$ for $z \geq 0$.